CORE PURE / POLAR COORDINATES

Integration of Polar Curves

Mark scheme 12 questions · 96 marks

Questions

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Question 12	(a) 4 (b) 6	10	16
	Total	**96**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

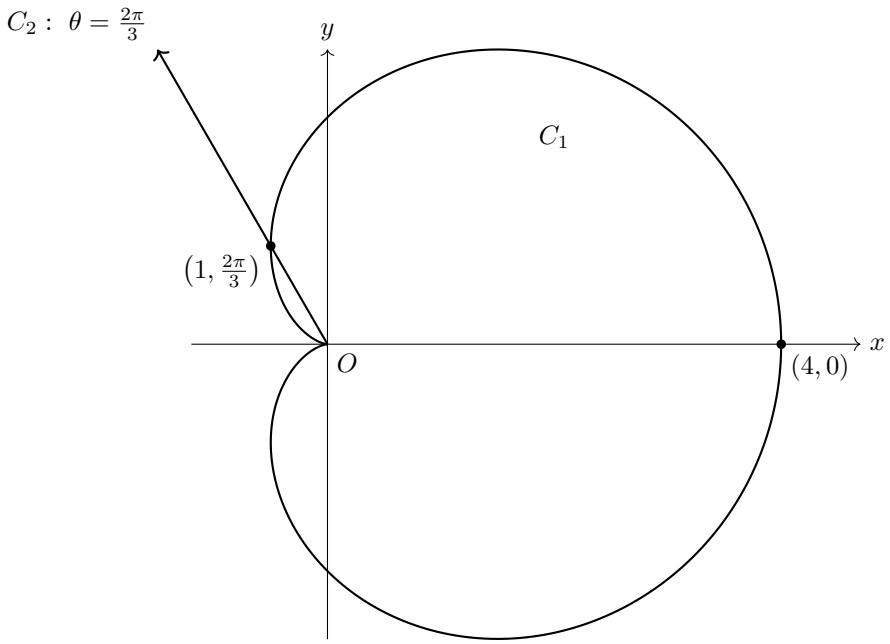
- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\sin 2\theta = -1 \Rightarrow \theta = \frac{3\pi}{4},\ \frac{7\pi}{4}$	**B1**	Both values, or awrt 2.36 and 5.50, with no others in $0 \leq \theta \leq 2\pi$.
		[1]	
1(b)	$\sin 2\theta = 1 \Rightarrow 2\theta = \frac{\pi}{2},\ \frac{5\pi}{2}$ $\Rightarrow \theta = \ldots$	**M1**	Sets $\sin 2\theta = 1$, or $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$, and solves for at least one value of θ.
	$\left(2, \frac{\pi}{4}\right)$ and $\left(2, \frac{5\pi}{4}\right)$	**A1**	Both points with $r = 2$; angles exact or awrt 0.785 and 3.93, with no others in the range.
		[2]	
1(c)	$A = \frac{1}{2}\int_0^{2\pi}(1+\sin 2\theta)^2\,\mathrm{d}\theta$	**B1**	Correct integral with limits, oe, such as $\int_0^{\pi}(1+\sin 2\theta)^2\,\mathrm{d}\theta$. Needs the polar-area factor $\frac{1}{2}$, allowing a compensating symmetry factor.
	$\frac{1}{2}\int_0^{2\pi}\left(\frac{3}{2}+2\sin 2\theta-\frac{1}{2}\cos 4\theta\right)\,\mathrm{d}\theta$ $= \frac{1}{2}\left[\frac{3}{2}\theta-\cos 2\theta-\frac{1}{8}\sin 4\theta\right]_0^{2\pi}$	**M1**	Expands, replaces $\sin^2 2\theta$ by $\pm\frac{1}{2}\pm\frac{1}{2}\cos 4\theta$ and integrates, with at least one trigonometric term integrated correctly. Condone coefficient slips.
	$= \frac{1}{2}\left(\text{“}3\pi - 1\text{”} - \text{“}(-1)\text{”}\right)$	**dM1**	Substitutes both limits into their integrated expression and subtracts.
	$= \frac{3\pi}{2}$	**A1**	cso. Withhold for an uncorrected wrong identity or integral, or a fortuitous answer; condone an explicitly corrected preliminary error.
		[4]	
1(d)	$r = 1 + 2\sin\theta\cos\theta \Rightarrow r = 1 + \frac{2xy}{r^2}$ $\Rightarrow r^3 = r^2 + 2xy$	**M1**	Uses $\sin 2\theta = 2\sin\theta\cos\theta$ with $x = r\cos\theta$, $y = r\sin\theta$ to reach an equation in r, x and y with no θ.
	$(x^2+y^2)^{3/2} = (x+y)^2$	**A1**	oe in x and y only, such as $(x^2+y^2)^3 = (x+y)^4$. A0 if r remains.
		[2]	
	Total for Question 1	**9**	

Question	Scheme	Marks	Notes
2(a)	$\cos 2\theta = \cos^2\theta - \sin^2\theta$ $= \left(\frac{x}{r}\right)^2 - \left(\frac{y}{r}\right)^2$	**M1**	Writes $\cos 2\theta$, or $r^2\cos 2\theta$, in terms of x, y and r using a double-angle identity with $x = r\cos\theta$, $y = r\sin\theta$. Condone sign slips.
	$\cos 2\theta = \frac{x^2-y^2}{r^2}$	**A1**	oe, such as $r^2\cos 2\theta = x^2 - y^2$.
	$r^2 = \frac{8(x^2-y^2)}{r^2}$ $\Rightarrow x^2 + y^2 = \frac{8(x^2-y^2)}{x^2+y^2}$	**dM1**	Substitutes their $\cos 2\theta$ in x, y and r into $r^2 = 8\cos 2\theta$ and replaces r^2 by $x^2 + y^2$.
	$(x^2+y^2)^2 = 8(x^2-y^2)$	**A1 AG**	Correct substituted equation with $r^2 = x^2 + y^2$, leading without error to the printed result; combined steps accepted.
		[4]	
2(b)	$\left(x^2 + (-y)^2\right)^2 = 8\left(x^2 - (-y)^2\right)$ $\left((-x)^2 + y^2\right)^2 = 8\left((-x)^2 - y^2\right)$	**M1**	Replaces one variable by its negative, or applies a general polar reflection test, and shows invariance for at least one axis.
	Both give $(x^2+y^2)^2 = 8(x^2-y^2)$, unchanged $\Rightarrow$ symmetric about both axes	**A1 AG**	Each result shown to be the original equation, with a conclusion that both axes are lines of symmetry.
		[2]	
2(c)	$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 8\cos 2\theta \, \mathrm{d}\theta$	**B1**	oe, such as $2 \times \frac{1}{2}\int_0^{\pi/4} 8\cos 2\theta \, \mathrm{d}\theta$ or limits $\frac{3\pi}{4}, \frac{5\pi}{4}$. Needs the polar-area factor $\frac{1}{2}$.
	$= 4\left[\frac{1}{2}\sin 2\theta\right]_{-\pi/4}^{\pi/4}$	**M1**	Integrates $\cos 2\theta$ to $k\sin 2\theta$, any $k \neq 0$.
	$= 2\left(\sin\frac{\pi}{2} - \sin\left(-\frac{\pi}{2}\right)\right)$	**dM1**	Substitutes their limits for one loop into their $k\sin 2\theta$ and subtracts. May be implied by a correct answer.
	$= 4$	**A1**	cso.
		[4]	
	Total for Question 2	**10**	

Question	Scheme	Marks	Notes
3(a)			
	Shape	**B1**	Cardioid symmetric about the initial line, with a cusp, not a loop or dimple, at the pole.
	Labelled point	**B1**	The intercept with the initial line marked as $(4, 0)$, or 4 at that point.
	Half-line	**B1**	Half-line from the pole into the second quadrant, at about $120°$, crossing the cardioid. Not a full line through the pole.
		[3]	
3(b)	$A = \frac{1}{2}\int_{2\pi/3}^{\pi} 4(1+\cos\theta)^2 \, \mathrm{d}\theta$	**B1**	Needs the polar-area factor $\frac{1}{2}$, a correct r^2 and limits $\frac{2\pi}{3}$ and π, seen or used later.
	$A = \int_{2\pi/3}^{\pi}(3 + 4\cos\theta + \cos 2\theta) \, \mathrm{d}\theta$ $= \left[3\theta + 4\sin\theta + \frac{1}{2}\sin 2\theta\right]_{2\pi/3}^{\pi}$	**M1**	Expands to three terms, uses $\cos^2\theta = \frac{1}{2}(1 \pm \cos 2\theta)$ and integrates to the form $a\theta + b\sin\theta + c\sin 2\theta$. Condone coefficient slips.
	$A = 3\pi - \left(2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{4}\right)$	**dM1**	Substitutes π and $\frac{2\pi}{3}$ into their integral of that form and subtracts.
	$A = \pi - \frac{7\sqrt{3}}{4}$	**A1**	Any exact equivalent, or awrt 0.111. cso.
		[4]	
	Total for Question 3	**7**	

Question	Scheme	Marks	Notes
4	$2+\sin 2\theta = \frac{5}{2} \Rightarrow \sin 2\theta = \frac{1}{2}$ $\Rightarrow \theta = \ldots$	**M1**	Equates the two expressions for r and solves for at least one value of θ.
	$\theta = \frac{\pi}{12},\ \frac{5\pi}{12}$	**A1**	Both values, exact or awrt 0.262 and 1.31.
	$\text{Area} = \frac{1}{2}\int_{\pi/12}^{5\pi/12} \left[(2+\sin 2\theta)^2 - \left(\frac{5}{2}\right)^2\right] \mathrm{d}\theta$	**B1**	Correct integral with correct limits, oe such as a separate circular sector subtracted. Needs the polar-area factor $\frac{1}{2}$.
	$\sin^2 2\theta = \frac{1-\cos 4\theta}{2}$ $\Rightarrow \left[-\frac{7\theta}{8} - \cos 2\theta - \frac{1}{16}\sin 4\theta\right]$	**M1**	Expands $(2+\sin 2\theta)^2$, replaces $\sin^2 2\theta$ by $\pm\frac{1}{2} \pm \frac{1}{2}\cos 4\theta$ and integrates, at least one trigonometric term correctly. Condone coefficient slips.
	$\left(-\frac{35\pi}{96} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{32}\right)$ $-\left(-\frac{7\pi}{96} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{32}\right)$	**dM1**	Substitutes their limits from the intersection into their integrated expression and subtracts.
	$\text{Area} = 0.924$	**A1**	awrt 0.924, cso. The exact value $\frac{17\sqrt{3}}{16} - \frac{7\pi}{24}$ alone scores A0.
	Total for Question 4	**6**	

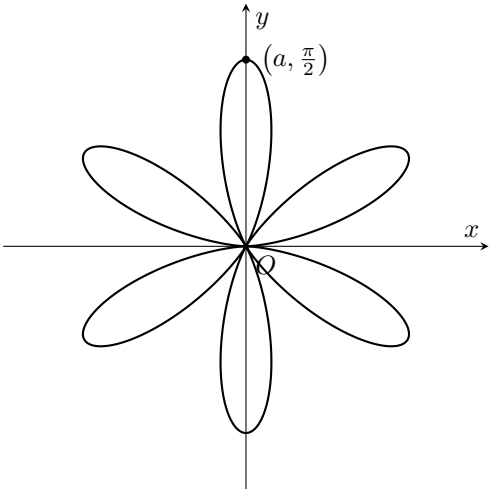
Question	Scheme	Marks	Notes
5(a)	A has polar coordinates $\left(3a, \frac{\pi}{4}\right)$	**B1**	
	B has polar coordinates $(2a, 0)$	**B1**	
		[2]	
5(b)	$r = a(2 + 2\sin\theta\cos\theta) = 2a + \frac{2axy}{r^2}$ $\Rightarrow (x^2 + y^2)^{3/2} = 2a(x^2 + y^2) + 2axy$	**M1**	Uses $\sin 2\theta = 2\sin\theta\cos\theta$, $x = r\cos\theta$, $y = r\sin\theta$ to reach an equation in x and y only. Condone slips in the powers of $x^2 + y^2$.
	Unchanged on swapping x and y $\theta = \frac{\pi}{4}$ is $y = x$, so C is symmetrical about it	**E1**	States that their correct Cartesian equation is unchanged on swapping x and y, identifies $\theta = \frac{\pi}{4}$ as $y = x$ and concludes.
	Alternative: Replacing θ by $\frac{\pi}{2} - \theta$		
	$r = a\left(2 + \sin 2\left(\frac{\pi}{2} - \theta\right)\right)$ $= a\left(2 + \sin(\pi - 2\theta)\right)$	**M1**	Replaces θ by $\frac{\pi}{2} - \theta$, or compares r at $\frac{\pi}{4} \pm \alpha$ for general α. Not for particular angles or the diagram alone.
	$\sin(\pi - 2\theta) = \sin 2\theta \Rightarrow r = a(2 + \sin 2\theta)$ Unchanged, so symmetrical about $\theta = \frac{\pi}{4}$	**E1**	Shows $\sin(\pi - 2\theta) = \sin 2\theta$, by identity or expansion, and concludes the equation is unchanged so C is symmetrical.
		[2]	
5(c)	$\sin 2\theta \geq 0 \Rightarrow 0 \leq \theta \leq \frac{\pi}{2}$ $\frac{1}{2}\int_0^{\pi/2} \left(a^2(2 + \sin 2\theta)^2 - 4a^2\right)\,\mathrm{d}\theta$	**B1**	oe, such as $\frac{1}{2}\int a^2(2 + \sin 2\theta)^2\,\mathrm{d}\theta - \pi a^2$. Limits $0, \frac{\pi}{2}$ or $-\pi, -\frac{\pi}{2}$, or both intervals. Needs the polar-area factor $\frac{1}{2}$.
	$\frac{a^2}{2}\int_0^{\pi/2} \left(4\sin 2\theta + \frac{1}{2} - \frac{1}{2}\cos 4\theta\right)\,\mathrm{d}\theta$ $= \frac{a^2}{2}\left[-2\cos 2\theta + \frac{\theta}{2} - \frac{1}{8}\sin 4\theta\right]_0^{\pi/2}$	**M1**	Expands $(2 + \sin 2\theta)^2$, uses $\sin^2 2\theta = \frac{1}{2}(1 \pm \cos 4\theta)$ and integrates, at least one term correct. Condone sign slips.
	$2 \times \frac{a^2}{2}\left(\left(2 + \frac{\pi}{4}\right) - (-2)\right)$	**dM1**	Substitutes their limits into their integrated expression and doubles, or adds the results for both intervals.
	$\frac{a^2(16+\pi)}{4}$	**A1**	Any exact equivalent, such as $a^2\left(4 + \frac{\pi}{4}\right)$; cso.
		[4]	
	Total for Question 5	**8**	

Question	Scheme	Marks	Notes
6(a)	$4a\cos\theta = 2a\sec\theta \Rightarrow \cos^2\theta = \ldots$	**M1**	Sets the two expressions for r equal and rearranges to $\cos^2\theta = \ldots$, $\cos\theta = \ldots$ or $\cos 2\theta = \ldots$. Condone slips.
	$\theta = \frac{\pi}{4}$	**A1**	Or awrt 0.785. A0 if other values are given and not rejected.
	$r = 4a\cos$ "$\frac{\pi}{4}$" $= \ldots$	**dM1**	Substitutes their θ into either equation to find r.
	P has polar coordinates $\left(2\sqrt{2}\,a,\ \frac{\pi}{4}\right)$	**A1**	$r = 2\sqrt{2}\,a$ oe or awrt $2.83a$, with $\theta = \frac{\pi}{4}$ or awrt 0.785; need not be written as a pair.
		[4]	
6(b)	$r^2 = 4ar\cos\theta \Rightarrow x^2 + y^2 = 4ax$	**M1**	Multiplies by r (or otherwise) and uses $r^2 = x^2 + y^2$, $r\cos\theta = x$ to reach a Cartesian equation. Condone slips.
	$(x-2a)^2 + y^2 = 4a^2$ Centre $(2a, 0)$, radius $2a$	**A1**	Correct centre and radius, stated or seen in the circle's equation.
	Line $x = 2a$ through the centre; P is $(2a, 2a)$ in Cartesian coordinates: quarter circle, $A = \frac{1}{4}\pi($"$2a$"$)^2$	**M1**	Identifies $r = 2a\sec\theta$ as $x = 2a$ through their centre and takes the region as a quarter of their circle's area.
	$A = \pi a^2$	**A1**	πa^2 only; a decimal multiple of a^2 scores A0.
	Alternative: Integration between the curves		
	$A = \frac{1}{2}\int_0^{\pi/4}(4a\cos\theta)^2\,\mathrm{d}\theta$ $\quad - \frac{1}{2}\int_0^{\pi/4}(2a\sec\theta)^2\,\mathrm{d}\theta$	**B1**	Needs the polar-area factor $\frac{1}{2}$, both squares and limits 0, $\frac{\pi}{4}$. Or the first integral minus $2a^2$, the area of triangle O, $(2a, 0)$, P.
	$8a^2\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\pi/4}$ $\quad - 2a^2\left[\tan\theta\right]_0^{\pi/4}$	**M1**	Uses $\cos^2\theta = \frac{1}{2}(1 \pm \cos 2\theta)$ and integrates to $p\theta + q\sin 2\theta$, and $\sec^2\theta$ to $\tan\theta$ where used. Condone coefficient and sign slips.
	$= 8a^2\left(\frac{\pi}{8} + \frac{1}{4}\right) - 2a^2(1)$	**dM1**	Substitutes $\frac{\pi}{4}$ and 0 into their integrated expression and subtracts.
	$= \pi a^2$	**A1**	cso. πa^2 only; a decimal multiple of a^2 scores A0.
		[4]	
	Total for Question 6	**8**	

Question	Scheme	Marks	Notes
7	$a(1+\cos\theta) = a \Rightarrow \cos\theta = 0$ $\Rightarrow \theta = \ldots$	**M1**	Sets the two expressions for r equal and solves for θ. Condone slips in rearranging.
	$\theta = \pm\frac{\pi}{2}$	**A1**	Both values, or $\frac{\pi}{2}$ with symmetry used later; may be implied by the limits.
	$A = \frac{1}{2}\int_{-\pi/2}^{\pi/2}\left([a(1+\cos\theta)]^2 - a^2\right)\,\mathrm{d}\theta$	**B1**	Or $2\times\frac{1}{2}\int_{0}^{\pi/2}$, or the cardioid sector minus the semicircle. Needs the polar-area factor $\frac{1}{2}$ and matching limits.
	$= \frac{a^2}{2}\int\left(2\cos\theta + \frac{1}{2} + \frac{1}{2}\cos 2\theta\right)\,\mathrm{d}\theta$ $= \frac{a^2}{2}\left[2\sin\theta + \frac{\theta}{2} + \frac{1}{4}\sin 2\theta\right]$	**M1**	Expands $(1+\cos\theta)^2$, uses $\cos^2\theta = \frac{1}{2}(1\pm\cos 2\theta)$ and integrates, at least one term correct. Condone a sign or coefficient slip.
	$\frac{a^2}{2}\left[\ldots\right]_{-\pi/2}^{\pi/2} = \frac{a^2}{2}\left(4+\frac{\pi}{2}\right)$	**dM1**	Substitutes their limits into their integrated expression and subtracts.
	$= 2a^2 + \frac{\pi a^2}{4}$	**A1**	oe, such as $\frac{a^2}{4}(8+\pi)$.
	"$2a^2 + \frac{\pi a^2}{4}$" $= 18 + \frac{9\pi}{4} \Rightarrow a^2 = \ldots$	**M1**	Equates their area, of the form ka^2 with k a non-zero constant, to the given value and makes a^2 the subject.
	$a = 3$	**A1**	A0 if $a = -3$ is also given and not rejected.
	Total for Question 7	**8**	

Question	Scheme	Marks	Notes
8	$a\sin 2\theta = a\cos\theta$ $\Rightarrow \cos\theta(2\sin\theta - 1) = 0 \Rightarrow \theta = \ldots$	**M1**	Equates the curves, uses $\sin 2\theta = 2\sin\theta\cos\theta$ and solves to $\sin\theta = \ldots$ or $\theta = \ldots$. Condone dividing by $\cos\theta$.
	$\theta = \frac{\pi}{6}$	**A1**	$\theta = \frac{\pi}{2}$ need not be seen.
	$A = \frac{1}{2}\int_0^{\pi/6} a^2\sin^2 2\theta\,\mathrm{d}\theta$ $+ \frac{1}{2}\int_{\pi/6}^{\pi/2} a^2\cos^2\theta\,\mathrm{d}\theta$	**B1**	Each integral needs its $\frac{1}{2}$, a^2, integrand and limits; the factor $\frac{1}{2}a^2$ may be taken outside both.
	Uses $\sin^2 2\theta = \frac{1-\cos 4\theta}{2}$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ and integrates	**M1**	Uses double-angle identities for both squares and integrates to the form $p\theta + q\sin k\theta$. Condone sign and coefficient slips.
	$\left[\frac{\theta}{2} - \frac{\sin 4\theta}{8}\right]_0^{\pi/6}$, $\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{\pi/6}^{\pi/2}$	**A1**	Both correct, oe; constant factors may be inside or outside.
	$\frac{1}{2}a^2\left(\text{“}\frac{\pi}{12} - \frac{\sqrt{3}}{16}\text{”} + \text{“}\frac{\pi}{6} - \frac{\sqrt{3}}{8}\text{”}\right)$	**dM1**	Substitutes the limits into both their integrated expressions, of the form $p\theta + q\sin k\theta$, and adds.
	$\frac{a^2(4\pi - 3\sqrt{3})}{32}$	**A1**	Any exact equivalent; cso.
	Total for Question 8	**7**	

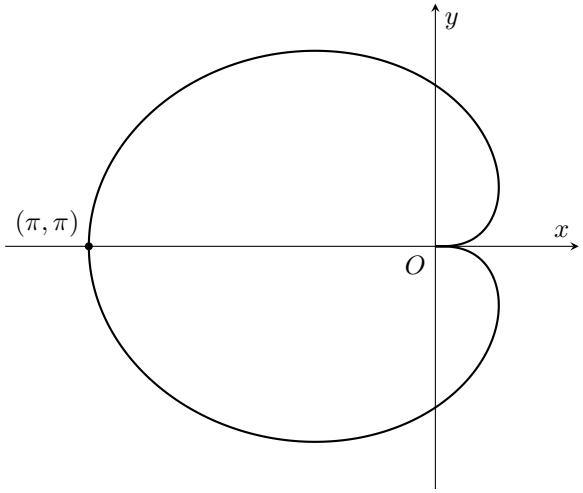
Question	Scheme	Marks	Notes
9(a)	$2a(1-\cos\theta) = a \Rightarrow \cos\theta = \ldots$	**M1**	Sets the two expressions for r equal and rearranges to $\cos\theta = k$. Condone sign slips.
	$\cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3},\ \frac{5\pi}{3}$	**A1**	Both values, exact or awrt 1.05 and 5.24. A0 for $-\frac{\pi}{3}$ in place of $\frac{5\pi}{3}$.
	$\left(a, \frac{\pi}{3}\right),\ \left(a, \frac{5\pi}{3}\right)$	**A1**	Both points as polar coordinates with $r = a$; θ as in the previous mark.
		[3]	
9(b)	$2 \times \frac{1}{2}\int_0^{\pi/3} 4a^2(1-\cos\theta)^2\,\mathrm{d}\theta$	**B1**	Correct integral for both cardioid sections: limits 0 to $\frac{\pi}{3}$ doubled, $-\frac{\pi}{3}$ to $\frac{\pi}{3}$, or separate integrals over 0 to $\frac{\pi}{3}$ and $\frac{5\pi}{3}$ to 2π.
	$\frac{1}{2}a^2\left(\frac{5\pi}{3} - \frac{\pi}{3}\right) = \frac{2\pi a^2}{3}$	**B1**	By integration or as a fraction of the circle's area.
	$4a^2\int\left(1 - 2\cos\theta + \frac{1+\cos 2\theta}{2}\right)\,\mathrm{d}\theta$ $= 4a^2\left[\frac{3}{2}\theta - 2\sin\theta + \frac{1}{4}\sin 2\theta\right]$	**M1**	Expands $(1-\cos\theta)^2$, uses $\cos^2\theta = \frac{1}{2}(1 \pm \cos 2\theta)$ and integrates, with at least one term correct. Condone slips in the constant factor.
	$4a^2\left(\frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8}\right) + \text{“}\frac{2\pi a^2}{3}\text{”}$	**dM1**	Substitutes limits 0 and $\frac{\pi}{3}$, or equivalent, into their integrated cardioid term and adds their sector area.
	$\frac{1}{6}a^2\left(16\pi - 21\sqrt{3}\right)$ $p = 16,\ q = -21$	**A1**	cso. $p = 16$, $q = -21$, or the area in the given form. Not for $\frac{8\pi a^2}{3} - \frac{7\sqrt{3}}{2}a^2$ alone.
		[5]	
Total for Question 9		**8**	

Question	Scheme	Marks	Notes
10(a)			
	Shape	**B1**	Six closed petals of equal size, each meeting the pole, with no part of the curve elsewhere.
	Labelled point	**B1**	Petals centred on $\theta = \frac{\pi}{6} + \frac{k\pi}{3}$, one along each direction of the y-axis, with $r = a$ marked at a tip.
		[2]	
10(b)	$\frac{1}{2}\int_0^{\pi/3} a^2 \sin^4 3\theta \, \mathrm{d}\theta$	**B1**	Limits for one petal, or 0 to 2π, or other limits with the matching multiple; needs $\frac{1}{2}$ and a^2.
	$\sin^4 3\theta = \left(\frac{1-\cos 6\theta}{2}\right)^2$ $= \frac{1}{4}\left(1 - 2\cos 6\theta + \frac{1+\cos 12\theta}{2}\right)$	**M1**	Uses a double-angle identity twice, or any valid method, to reach a constant and terms in $\cos 6\theta$ and $\cos 12\theta$; condone coefficient slips.
	$\sin^4 3\theta = \frac{1}{8}(3 - 4\cos 6\theta + \cos 12\theta)$	**A1**	oe, including within the integrand.
	$\frac{a^2}{16}\left[3\theta - \frac{2}{3}\sin 6\theta + \frac{1}{12}\sin 12\theta\right]$	**M1**	Integrates their $p + q\cos 6\theta + s\cos 12\theta$ to the form $\alpha\theta + \beta\sin 6\theta + \gamma\sin 12\theta$.
	$6 \times \frac{a^2}{16}\left[3 \times \frac{\pi}{3} - 0 + 0\right]$	**dM1**	Substitutes their limits and multiplies by the number of copies those limits need (6 for one petal, 1 for 0 to 2π).
	$\frac{3\pi a^2}{8}$	**A1**	cso, oe.
		[6]	
	Total for Question 10	**8**	

Question	Scheme	Marks	Notes
11	$(1+2\sin\theta)\sin\theta = 1$ $\Rightarrow 2\sin^2\theta + \sin\theta - 1 = 0$ $\Rightarrow \sin\theta = \ldots$	**M1**	Sets $r\sin\theta = 1$ with $r = 1+2\sin\theta$, forms a three-term quadratic in $\sin\theta$ and solves it by any means, calculator allowed.
	$\theta = \frac{\pi}{6}$	**A1**	Condone $\sin\theta = -1$ not explicitly rejected.
	$\frac{1}{2}\int_{\pi/6}^{\pi/2}(1+2\sin\theta)^2\,\mathrm{d}\theta$	**B1**	Limits may first be seen when substituted.
	$= \frac{1}{2}\int(3+4\sin\theta-2\cos2\theta)\,\mathrm{d}\theta$ $= \frac{1}{2}[3\theta - 4\cos\theta - \sin2\theta]$	**M1**	Expands $(1+2\sin\theta)^2$ to three terms, uses $\sin^2\theta = \frac{1}{2}(1\pm\cos2\theta)$ and integrates, at least one trigonometric term correct. Condone coefficient slips.
	$\frac{1}{2}\left[\frac{3\pi}{2} - \left(\frac{\pi}{2} - 2\sqrt{3} - \frac{\sqrt{3}}{2}\right)\right]$ $\left[= \frac{\pi}{2} + \frac{5\sqrt{3}}{4}\right]$	**dM1**	Substitutes $\frac{\pi}{2}$ and their θ at A into their integrated expression and subtracts, upper minus lower.
	Triangle $ONA = \frac{1}{2}\times 1\times\sqrt{3} = \frac{\sqrt{3}}{2}$	**B1**	
	Area $= \frac{\pi}{2} + \frac{5\sqrt{3}}{4} - \frac{\sqrt{3}}{2}$ $= \frac{\pi}{2} + \frac{3\sqrt{3}}{4}$	**A1**	cso. Any exact equivalent, such as $\frac{2\pi+3\sqrt{3}}{4}$.
	Alternative: Area between the curve and $r = \operatorname{cosec}\theta$		
	$(1+2\sin\theta)\sin\theta = 1$ $\Rightarrow 2\sin^2\theta + \sin\theta - 1 = 0$ $\Rightarrow \sin\theta = \ldots$	**M1**	Sets $r\sin\theta = 1$ with $r = 1+2\sin\theta$, forms a three-term quadratic in $\sin\theta$ and solves it by any means, calculator allowed.
	$\theta = \frac{\pi}{6}$	**A1**	Condone $\sin\theta = -1$ not explicitly rejected.
	$r\sin\theta = 1 \Rightarrow r = \operatorname{cosec}\theta$	**B1**	Or $r = \frac{1}{\sin\theta}$.
	Area $= \frac{1}{2}\int_{\pi/6}^{\pi/2}\left[(1+2\sin\theta)^2 - \operatorname{cosec}^2\theta\right]\,\mathrm{d}\theta$ $= \frac{1}{2}\int(3+4\sin\theta-2\cos2\theta-\operatorname{cosec}^2\theta)\,\mathrm{d}\theta$	**M1**	Uses $\frac{1}{2}\int$ of $(1+2\sin\theta)^2$ minus the square of their r for AN, expands, uses $\sin^2\theta = \frac{1}{2}(1\pm\cos2\theta)$ and integrates at least one trigonometric term correctly.
	$= \frac{1}{2}[3\theta - 4\cos\theta - \sin2\theta + \cot\theta]$	**A1**	All four terms correct; the $\frac{1}{2}$ may be applied later.
	$\frac{1}{2}\left[\frac{3\pi}{2} - \left(\frac{\pi}{2} - \frac{5\sqrt{3}}{2} + \sqrt{3}\right)\right]$	**dM1**	Substitutes $\frac{\pi}{2}$ and their θ at A into their integrated expression and subtracts, upper minus lower.

Question 11 continued

Question	Scheme	Marks	Notes
	$= \frac{\pi}{2} + \frac{3\sqrt{3}}{4}$	**A1**	cso. Any exact equivalent, such as $\frac{2\pi+3\sqrt{3}}{4}$.
	Total for Question 11	**7**	

Question	Scheme	Marks	Notes
12(a)			
	Shape	**B1**	One closed loop through the pole, lying mainly on the $\theta = \pi$ side of it, symmetric about the initial line.
	Intercepts	**B1**	Both ends of the curve at the pole. Condone the tangent there not along the initial line.
	$\theta = \pi$: $r^2 = \frac{\pi^2 - 0}{1+0} \Rightarrow r = \ldots$	**M1**	Substitutes $\theta = \pi$ into the equation to find r. No justification that this is the furthest point is needed.
	(π, π)	**A1**	Or $r = \pi$ and $\theta = \pi$ stated.
		[4]	
12(b)	$A = \frac{1}{2}\int_0^{2\pi} \frac{\pi^2 - (\theta-\pi)^2}{1+(\theta-\pi)^2}\,\mathrm{d}\theta$	**B1**	$\frac{1}{2}$, r^2 substituted and limits 0, 2π; or $\int_0^{\pi}$ of r^2 by symmetry.
	$u = \theta - \pi$: $\frac{\pi^2 - u^2}{1+u^2} = -1 + \frac{\ldots}{1+u^2}$	**M1**	Replaces $\theta - \pi$ by u and writes the integrand as $a + \frac{b}{1+u^2}$, by division or otherwise. Condone wrong a and b.
	$A = \frac{1}{2}\int_{-\pi}^{\pi} \left(-1 + \frac{\pi^2+1}{1+u^2}\right)\,\mathrm{d}u$	**A1**	Correct integrand with limits $\pm\pi$, oe, such as $\int_0^{\pi}$ without the $\frac{1}{2}$. Limits 0, 2π with a u integrand is A0.
	$= \frac{1}{2}\left[-u + (\pi^2+1)\arctan u\right]_{-\pi}^{\pi}$	**M1**	Integrates their $a + \frac{b}{1+u^2}$ to $au + b\arctan u$. Condone wrong a and b.
	$\frac{1}{2}\left[\left(-\pi + (\pi^2+1)\arctan\pi\right)\right.$ $\left. - \left(\pi - (\pi^2+1)\arctan\pi\right)\right]$	**dM1**	Substitutes the limits of their correctly transformed full or half interval into their antiderivative, subtracts and applies any required symmetry factor.

Question 12 continued

Question	Scheme	Marks	Notes
	$A = (\pi^2 + 1)\arctan\pi - \pi$	**A1**	oe, such as $(\pi^2 + 1)\arctan\pi - \pi$. cso.
		[6]	
	Total for Question 12	**10**	